
Second-Order Sharpness-Aware Minimization for Heterogeneous Federated Learning

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Abstract

The goal of any machine learning system is to generalize effectively to unseen data. In federated learning (FL), this challenge is amplified by data privacy constraints and heterogeneous, non-IID client distributions, where data varies significantly across decentralized clients. To address this, we propose a second-order perturbation-based approach to Sharpness-Aware Minimization (SAM) tailored for federated learning. Our method leverages the maximum eigenvector and eigenvalue of the Hessian, computed via power iteration, alongside trace-based sharpness regularization using Hutchinson’s method. This enables more precise control over local sharpness, promoting convergence to flatter minima that generalize better. We evaluate our method on vision classification tasks using CIFAR-10 and CIFAR-100 under synthetic heterogeneous (Dirichlet label-skew) splits. Our variants consistently outperform FedAvg and first-order FedSAM in test accuracy, by up to 5.77 points on CIFAR-10, and reach 50% accuracy up to 100 communication rounds earlier, while converging to measurably flatter minima. A key limitation in scaling second-order methods is their computational cost, which we measure at $3.5\text{--}12.5\times$ that of FedAvg per round. To mitigate this, we propose sparse second-order perturbations interleaved with first-order updates, second-order switching in later training rounds, and a hybrid with server-side global sharpness minimization (FedGloSS); their empirical evaluation is left to future work. Our code is available at https://github.com/muhammad-ahmad-1/fedsam_hessian.

1 Introduction

In typical machine learning scenarios, models are trained on a dataset with the assumption that training and test data come from the same underlying distribution. However, real-world data often violates this assumption, as data is diverse, noisy, and prone to significant shifts, creating a need for models that generalize to unseen distributions [Tran et al., 2022, Wang et al., 2022a]. Federated learning (FL), a privacy-preserving distributed learning paradigm [McMahan et al., 2017], introduces a further challenge: data heterogeneity across clients, where client data is non-IID. This adds an additional layer of distribution shift between clients, leading to issues like client drift and poor generalization of the global model.

Generalization-enhancing methods such as Sharpness-Aware Minimization (SAM) [Foret et al., 2020] have shown promise by encouraging models to converge to flatter regions of the loss landscape that are typically more robust. While existing methods like FedSAM [Qu et al., 2022, Caldarola et al., 2022] have adapted SAM for federated learning, they primarily rely on first-order approximations of sharpness and do not leverage the full geometry of the loss surface. Existing methods like FedAvg and FedProx [Li et al., 2020] struggle with client drift, and first-order sharpness-aware techniques

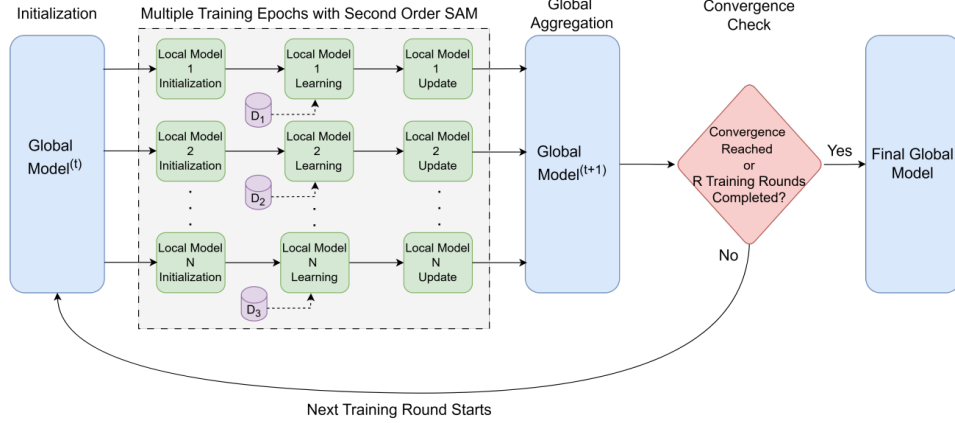


Figure 1: Overview. In each communication round, the global model is broadcast; clients train locally for multiple epochs with a second-order SAM variant on their private data; the server aggregates the local models into the next global model until convergence or a fixed number of rounds.

rely on approximate measures of sharpness, neglecting richer second-order information that could guide optimization more effectively.

This project goes a step further by incorporating second-order information, specifically, leading eigenvectors and eigenvalues of the Hessian and trace-based approximations, to better quantify and minimize sharpness during local client training (Figure 1). Our contributions are:

- **Second-order FedSAM variants.** We devise four client-side variants, FEDSAM_HESSIAN, FEDSAM_HESSIAN_ORTH, FEDSAM_TRACE, and FEDSAM_EIGEN_TRACE, which use power-iteration estimates of the Hessian’s leading eigenvector and Hutchinson estimates of its trace to perturb and regularize local training (Section 3).
- **Empirical evaluation.** On CIFAR-10 and CIFAR-100 under Dirichlet label skew, with full and partial participation, the variants consistently outperform FedAvg and first-order FedSAM in test accuracy, converge in fewer rounds, produce better-aligned local updates, and reach flatter minima as measured by the Hessian’s top eigenvalue and trace (Section 5).
- **Cost characterization and efficient extensions.** We measure the per-round overhead of each variant, and propose interleaved and switched sparse second-order schedules, as well as a hybrid with server-side global sharpness minimization (FedGloSS [Caldarola et al., 2024]), as routes to making such methods feasible in resource-constrained federated environments.
- **Open-source framework.** A modular federated training codebase with a one-to-one server/client class correspondence for each algorithm, supporting reproducibility and extension.

2 Background and Related Work

Federated learning and data heterogeneity. In FL, a global model is trained collaboratively across multiple decentralized clients (e.g., mobile devices, edge nodes, or organizations), each holding a local dataset that never leaves the client. FedAvg [McMahan et al., 2017] performs multiple local epochs on each client before aggregation:

$$\theta_{t+1} \leftarrow \sum_{k \in S_t} \frac{n_k}{n} \theta_{t+1}^k, \quad (1)$$

where S_t is the subset of m randomly selected clients in round t , n_k is the size of client k ’s dataset, $n = \sum_{k \in S_t} n_k$, and θ_{t+1}^k are client k ’s parameters after E local epochs. FedSGD is the special case of a single local gradient step with full participation. FedAvg works well when client distributions are similar, but in practice client data is often non-IID, manifesting as label skew, data imbalance, or domain shift. Local updates then diverge from each other and from the global objective (client drift,

Figure 7b in Appendix B), causing slow convergence and poor generalization [Karimireddy et al., 2020]; the generalization issues of the centralized case are inherited as well.

Generalization methods in federated learning. Methods addressing heterogeneity and domain shift in FL are largely extensions of centralized domain-generalization methods [Wang et al., 2022a] and fall into four broad categories [Li et al., 2023]: (1) domain alignment, e.g., adversarial alignment [Peng et al., 2019, Wang et al., 2022b] and model-level contrastive learning [Li et al., 2021]; (2) data manipulation, e.g., augmentation [Duan et al., 2019], style transfer [Shenaj et al., 2023, Chen et al., 2023], and synthetic data generation [Zhang and Jiang, 2022]; (3) learning strategies such as representation learning, meta-learning, knowledge distillation, and regularization; and (4) aggregation optimization, e.g., fairness-aware, cluster- and layer-aware, or gradient-aligning aggregation. Proximal regularization [Li et al., 2020] and control variates [Karimireddy et al., 2020] limit client drift directly, and initialization from models pre-trained on proxy data also helps in heterogeneous settings [Nguyen et al., 2022]. This work modifies client-side training through sharpness-aware optimization.

Sharpness-aware minimization. SAM [Foret et al., 2020] seeks parameters that lie in neighborhoods with uniformly low loss, via a min-max objective over a perturbation radius ρ :

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon). \quad (2)$$

As the inner maximization is infeasible in high dimensions, SAM uses a first-order approximation, which yields the direction of maximum ascent:

$$\hat{\epsilon}(\theta) \triangleq \arg \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon) \approx \arg \max_{\|\epsilon\|_2 \leq \rho} \epsilon^\top \nabla_{\theta} \mathcal{L}(\theta) = \rho \frac{\nabla_{\theta} \mathcal{L}(\theta)}{\|\nabla_{\theta} \mathcal{L}(\theta)\|_2}. \quad (3)$$

Each step therefore (1) computes the adversarial perturbation $\hat{\epsilon}$ within the ρ -ball, and (2) updates the parameters using the gradient evaluated at $\theta + \hat{\epsilon}$ [Foret et al., 2020]. FedSAM and MoFedSAM [Qu et al., 2022, Caldarola et al., 2022] apply SAM in local client training, with the server aggregating the resulting parameters; related flatness-seeking approaches include weight averaging [Izmailov et al., 2018, Cha et al., 2021] and FedSMOO [Sun et al., 2023].

Second-order sharpness-aware minimization. Second-order methods leverage the Hessian to capture the curvature of the loss landscape more precisely. SASSHA [Shin et al., 2025] explicitly reduces sharpness through stable Hessian approximations with lazy Hessian updates, ensuring both efficiency and robustness. Eigen-SAM [Luo et al., 2024] addresses the fact that the SAM perturbation may not align well with the top eigenvector of the Hessian, and regularizes the top eigenvalue by aligning the perturbation with the leading eigenvector. Liu et al. [2022] penalize the trace of the Hessian, motivated by generalization bounds, using the Hutchinson estimator accelerated by a dropout scheme. These methods target centralized learning; our work extends them to the federated scenario and adapts them to its computational budget.

Global sharpness in federated learning. Recent federated work instead focuses on reaching globally flatter minima, since local sharpness at the client level does not account for the global loss landscape. FedLESAM [Fan et al., 2024] estimates the global perturbation locally from the difference between successive global models, requiring a single backpropagation per iteration. FedGloSS [Caldarola et al., 2024] moves sharpness minimization to the server, approximating the global perturbation from historical global pseudo-gradients without additional client computation. Because FedGloSS is purely server-side, it can be combined with our client-side second-order methods (Section 3.4); integration with the client-side FedLESAM would require significant modifications and is left to future work.

3 Method

Our methodology is driven by the goal of achieving flatter minima under non-IID data while remaining computationally feasible on clients. The Hessian $H = \nabla_{\theta}^2 \mathcal{L}(\theta) \in \mathbb{R}^{d \times d}$ captures the curvature of the loss landscape: its largest eigenvalue indicates the maximum curvature, and its trace the average curvature. Forming H is infeasible for neural networks ($O(d^2)$ compute and memory), so we estimate both quantities from Hessian-vector products, $Hv = \nabla_{\theta}(\nabla_{\theta} \mathcal{L}(\theta)^\top v)$, computed by automatic differentiation with two backward passes.

3.1 Hessian estimators

Power iteration for the principal eigenvector. Starting from a random unit vector v_0 (sampled uniformly on the unit sphere to ensure a non-zero component along the dominant eigenvector), each iteration computes $v_k = H v_{k-1} / \|H v_{k-1}\|_2$ and the Rayleigh quotient $\lambda_k = v_k^\top H v_k$, stopping when λ_k stabilizes or after a fixed number of iterations. The method converges to the largest eigenvalue and its eigenvector provided the eigenvalue is distinct and v_0 has a non-zero projection onto the dominant eigenvector. Neural network Hessians often exhibit a large eigenvalue gap, so convergence is typically rapid; we limit the number of iterations to at most 5–10, reducing the cost from $O(d^2)$ to $O(kd)$ for k iterations. Each client performs power iteration locally on its own loss.

Hutchinson’s trace estimator. For symmetric H , $\text{Tr}(H) = \mathbb{E}[z^\top H z]$ for any random z with zero mean and identity covariance, such as Rademacher ($z_i \in \{-1, 1\}$) or standard normal vectors. We approximate

$$\text{Tr}(H) \approx \frac{1}{m} \sum_{i=1}^m z_i^\top H z_i, \quad (4)$$

at a cost of $O(md)$. Typical choices are $m \approx 5$ –10; we experiment with lower values of around 2–5. A high trace indicates a sharp landscape with high average curvature, while a low trace indicates a flatter region associated with better generalization. Using the trace as a loss penalty, however, requires retaining the computational graph of each Hessian-vector product for backpropagation, which is significantly more expensive than the eigenvector perturbation.

3.2 Second-order FedSAM variants

Each client computes its perturbation locally, updates its model on the SAM objective $\min_\theta \mathcal{L}(\theta + \epsilon^*)$, and sends the updated parameters to the server for aggregation by Eq. (1). The variants differ in how ϵ^* and the local objective are formed (Algorithm 1).

FEDSAM_HESSIAN. The baseline approach adds a perturbation along the Hessian’s leading eigenvector $v_{\max}$, the direction of maximum curvature, to the first-order perturbation:

$$\epsilon^* = \rho \left(\frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|_2} + \hat{v} \right), \quad \hat{v} = \frac{v_{\max}}{\|v_{\max}\|_2}. \quad (5)$$

FEDSAM_HESSIAN_ORTH. Following Luo et al. [2024], this variant adds a corrective term orthogonal to the first-order perturbation:

$$\epsilon^* = \frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|_2} + \alpha \text{sign}(\langle \nabla_\theta \mathcal{L}(\theta), \hat{v} \rangle) \hat{v}^\perp, \quad (6)$$

where $\hat{v}^\perp$ is the component of $\hat{v}$ orthogonal to the gradient, obtained by Gram–Schmidt orthogonalization, the sign term orients it consistently with the gradient’s projection onto $\hat{v}$, and α is a small scaling factor (e.g., 0.1–0.5). This addresses the limitation of FEDSAM_HESSIAN in regions where the gradient and eigenvector are misaligned, capturing curvature in directions orthogonal to the gradient while keeping the optimization stable.

FEDSAM_TRACE. This variant complements first-order SAM with a regularizer on the average curvature, applied after the ascent step:

$$\tilde{\mathcal{L}}(\theta) = \mathcal{L}(\theta + \epsilon^*) + \lambda \text{Tr}(H(\theta + \epsilon^*)), \quad (7)$$

with the trace estimated by Hutchinson’s method. This aligns with Liu et al. [2022], who highlight the importance of average curvature, and becomes particularly relevant in non-IID settings where client-specific loss landscapes exhibit varying degrees of sharpness.

FEDSAM_EIGEN_TRACE. This hybrid combines the second-order ascent step of FEDSAM_HESSIAN_ORTH (Eq. 6) with the trace regularizer (Eq. 7). The eigenvector perturbation avoids sharp minima in the direction of maximum curvature, while the trace term promotes overall flatness across all directions.

Algorithm 1 Local client update with second-order SAM (one mini-batch step)

Require: parameters θ , mini-batch B , radius ρ , iterations k , variant

- 1: $g \leftarrow \nabla_{\theta} \mathcal{L}_B(\theta)$
 - 2: $\hat{v} \leftarrow \text{POWERITERATION}(H_B(\theta), k)$ ▷ Hessian-vector products only
 - 3: Form ϵ^* by Eq. (5) (HESSIAN) or Eq. (6) (HESSIAN_ORTH, EIGEN_TRACE); first-order Eq. (3) for TRACE
 - 4: $\ell \leftarrow \mathcal{L}_B(\theta + \epsilon^*)$
 - 5: **if** variant uses trace **then**
 - 6: $\ell \leftarrow \ell + \lambda \cdot \text{HUTCHINSON}(H_B(\theta + \epsilon^*), m)$ ▷ graph retained for backprop
 - 7: **end if**
 - 8: $\theta \leftarrow \theta - \eta \nabla_{\theta} \ell$ ▷ gradient evaluated at the perturbed point
-

3.3 Compute-efficient variants

FEDSAM_HESSIAN_INTERLEAVED. A strawman solution to meet a computational budget: within each local epoch, the second-order perturbation is applied only every k iterations, and the remaining iterations follow first-order SAM. This aligns with SASSHA [Shin et al., 2025], which similarly updates the Hessian lazily.

FEDSAM_HESSIAN_SWITCH. Motivated by the observation that the sharpness mitigation of first-order methods diminishes in later training rounds, this variant runs first-order SAM until a switching round T (a hyperparameter, e.g., 70% of total rounds) and FEDSAM_HESSIAN thereafter, applying curvature information when it matters most while reducing the number of second-order steps.

3.4 Hybrid with server-side global sharpness minimization

Because our methods act purely on clients, they are orthogonal to server-side improvements. We integrate them with FedGloSS [Caldarola et al., 2024], which applies SAM to the global objective $F(w) = \frac{1}{|C|} \sum_{k \in C} F_k(w)$ on the server, approximating the unavailable global perturbation by the previous round’s pseudo-gradient. Clients run a second-order variant (e.g., FEDSAM_EIGEN_TRACE or FEDSAM_HESSIAN_SWITCH) and send only their parameters; the server perturbs the global model, aggregates, and applies FedGloSS’s ADMM-based consistency update. The pipeline aims at minima that are both globally and locally flat without additional communication beyond our second-order SAM requirements. The full update equations are given in Appendix C. The hybrid is formulated and implemented; its empirical evaluation is future work.

4 Experimental Setup

Datasets and heterogeneity. We use MNIST for an initial toy problem, and CIFAR-10 and CIFAR-100 as the main benchmarks; CIFAR-100 tests fine-grained classification over 100 classes. Heterogeneity is synthesized through label skew: client partitions are drawn from a Dirichlet distribution whose concentration parameter α controls the degree of non-IIDness (lower values give more pronounced heterogeneity).

Model. We use a SimpleCNN with three convolutional layers (32, 64, 128 filters, each followed by ReLU and max-pooling) and fully connected layers (Figure 9, Appendix B). Its shallow depth keeps iterations quick, which is crucial for evaluating comparatively expensive second-order methods and isolating their effect.

Configurations and baselines. We evaluate FedAvg, first-order FedSAM, and the four second-order variants under three settings: full client participation with 35 and 100 rounds ($\alpha = 0.3$), and 10% partial participation with 800 rounds ($\alpha = 0.1$). Hyperparameters are listed in Table 1. We report test accuracy, convergence speed (rounds to reach 50% accuracy), wall time per round, the alignment of local updates (cosine similarity and variance of updates), and sharpness (the Hessian’s top eigenvalue and trace). Computing resources were provided by the Center for Urban Informatics, Technology, and Policy (CITY) at LUMS (Intel Core i9, NVIDIA RTX 3080 Ti).

Table 1: Hyperparameters (defaults of the released code).

Clients	10 (full participation); 100 (10% participation, 800 rounds)
Local optimizer	SGD, learning rate 5×10^{-3} , batch size 64, weight decay 10^{-4}
Local epochs	5 (3 for the wall-time measurements in Table 3)
SAM radius ρ	0.01
Power iterations / Hutchinson samples	5 (tolerance 10^{-2})
Orthogonal scale α (Eq. 6)	0.1
Trace weight λ (Eq. 7)	0.05

Table 2: Test accuracy (%) of the final global model. Full participation uses 10 clients with $\alpha = 0.3$; partial participation uses 10% of clients per round with $\alpha = 0.1$. Best in **bold**.

Algorithm	Full participation, $\alpha = 0.3$			10% participation, $\alpha = 0.1$	
	C10, 35 rds	C10, 100 rds	C100, 100 rds	C10, 800 rds	C100, 800 rds
FedAvg	55.43	70.45	37.82	54.45	33.42
FedSAM	55.91	70.66	40.25	54.01	33.49
FEDSAM_HESSIAN	59.24	74.68	40.75	56.14	36.34
FEDSAM_HESSIAN_ORTH	59.26	74.97	42.12	56.08	36.28
FEDSAM_TRACE	56.76	74.35	41.68	54.84	38.31
FEDSAM_EIGEN_TRACE	61.20	74.74	42.41	55.00	36.54

5 Results

Two preliminary experiments (Appendix A) motivated this setup. FedProx achieved slightly lower variance of updates than FedAvg across heterogeneity levels without translating into higher test accuracy, suggesting that update alignment alone is insufficient in non-IID settings. On a five-client MNIST toy problem, FEDSAM_HESSIAN outperformed FedSAM and converged in fewer rounds, but at roughly $10\times$ its cost with 10 power iterations.

Accuracy. Table 2 shows that all four second-order variants outperform FedAvg and FedSAM in every setting. FEDSAM_HESSIAN and FEDSAM_HESSIAN_ORTH are consistently strong; FEDSAM_EIGEN_TRACE achieves the highest accuracy on CIFAR-10 at 35 rounds (61.20%, +5.77 over FedAvg) and on CIFAR-100 at 100 rounds (42.41%, +4.59), while FEDSAM_TRACE is best in the high-heterogeneity, partial-participation setting on CIFAR-100 (38.31%, +4.89). The gains are largest in the short full-participation regime on CIFAR-10 and on CIFAR-100; on CIFAR-10 with 800 rounds and 10% participation they are smaller (up to +1.69).

Convergence and cost. FEDSAM_HESSIAN and FEDSAM_HESSIAN_ORTH reach 50% accuracy in 243–244 rounds versus 341–343 for FedAvg and FedSAM (Table 3), about 100 rounds earlier, reflecting the effectiveness of second-order perturbations in navigating the loss landscape. The trace-based variants also need fewer rounds (280–282) but are $9\text{--}12.5\times$ slower per round than FedAvg, due to backpropagating the Hessian trace; in wall-clock terms they therefore converge more slowly than FedAvg despite their higher final accuracy. The eigenvector-based variants cost $3.5\times$ FedAvg per round.

Alignment of local updates. The second-order variants achieve higher pairwise and global cosine similarity of local updates and lower discrepancy from the global update than FedAvg and first-order FedSAM (Figure 2), indicating reduced inter-client conflict and greater global-model consistency under non-IID data.

Sharpness. Both sharpness metrics grow steadily for FedAvg and FedSAM but remain low for the second-order variants (Figure 3). FEDSAM_EIGEN_TRACE reduces sharpness the most, followed by FEDSAM_TRACE, owing to their explicit regularization of average curvature; this aligns with the improved test accuracy of these variants.

Table 3: Convergence speed and cost on CIFAR-10 (800 rounds, 10% participation, $\alpha = 0.1$). Convergence is the number of rounds to reach 50% test accuracy; time is wall time per round for 3 local epochs, as logged.

Algorithm	Time / round	Slowdown vs. FedAvg	Rounds to 50%
FedAvg	~ 2	$1\times$	343
FedSAM	~ 3.35	$1.675\times$	341
FEDSAM_HESSIAN	~ 7	$3.5\times$	243
FEDSAM_HESSIAN_ORTH	~ 7	$3.5\times$	244
FEDSAM_TRACE	~ 18	$9\times$	282
FEDSAM_EIGEN_TRACE	~ 25	$12.5\times$	280

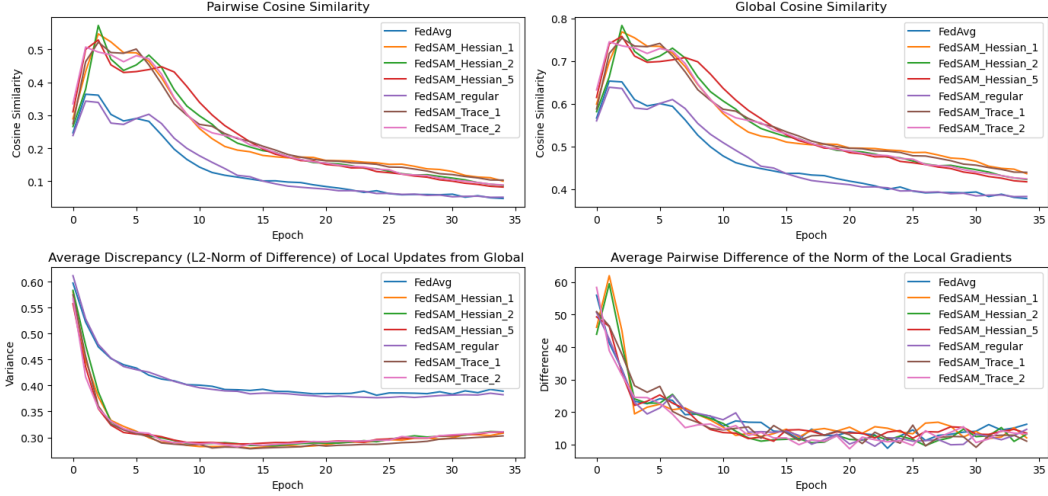


Figure 2: Alignment of local updates on CIFAR-10 over training: pairwise and global cosine similarity of local updates (top), average discrepancy of local updates from the global update, and average pairwise difference of local gradient norms (bottom). Suffixes denote the number of Hessian iterations; FedSAM_regular is first-order FedSAM.

6 Discussion, Limitations, and Future Work

The results show that second-order curvature information, both the leading eigenvector and the trace, guides local optimization toward flatter minima and improves generalization of the global model under label skew. The central trade-off is computational: the eigenvector-based variants balance accuracy gains with a $3.5\times$ per-round overhead, while the trace-based variants yield the flattest minima at $9\text{--}12.5\times$. The practical deployment of these variants is therefore constrained by their computational demands.

Several directions remain open. (i) *Efficiency*: the interleaved and switched schedules (Section 3.3) are implemented but not yet evaluated, and fixed switching and interleaving hyperparameters call for adaptive triggers based on lightweight sharpness proxies (e.g., gradient-norm ratios or cheap eigenvalue estimates). (ii) *Global flatness*: the FedGloSS hybrid (Section 3.4) requires empirical evaluation under realistic network constraints and partial participation, where global sharpness reduction could offset the loss from sparser local second-order steps. (iii) *Scale and shift*: extending the evaluation to deeper architectures (ResNet, ViT), to out-of-distribution benchmarks with domain skew (PACS and OfficeHome, via hold-out domains), and to extreme heterogeneity (a single class or domain per client), where sharpness-seeking alone is expected to be insufficient. (iv) *Broader applicability*: privacy-preserving Hessian approximations and non-vision tasks.

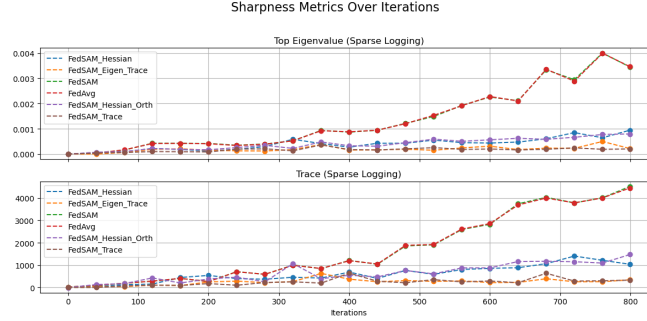


Figure 3: Sharpness over training: the Hessian’s top eigenvalue (top) and trace (bottom).

7 Conclusion

We developed and evaluated a suite of second-order sharpness-aware minimization methods for federated learning in non-IID, heterogeneous client environments. Using power iteration and Hutchinson’s method, both built on Hessian-vector products, our FedSAM variants guide local updates toward flatter minima and consistently outperform FedAvg and first-order FedSAM on CIFAR-10 and CIFAR-100, with up to a 5.77-point gain in test accuracy and up to 100 fewer communication rounds to reach 50% accuracy. These gains come at a $3.5\text{--}12.5\times$ per-round cost, which motivates the sparse schedules and server-side hybrid we propose as the next steps toward practical second-order federated optimization.

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A Preliminary Experiments

FedAvg and FedProx under varying heterogeneity. Ten clients were trained on CIFAR-10 splits with Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$ (higher α means lower heterogeneity), using a SimpleCNN with more filters per block than in the main experiments, which leads to higher accuracy (Figures 4–5). FedProx achieves slightly lower variance of updates across clients than FedAvg, attributed to its proximal term that keeps local updates close to the global model. This does not consistently translate into higher test accuracy, with FedAvg occasionally outperforming FedProx under high heterogeneity. FedProx was therefore not retained as a core baseline.

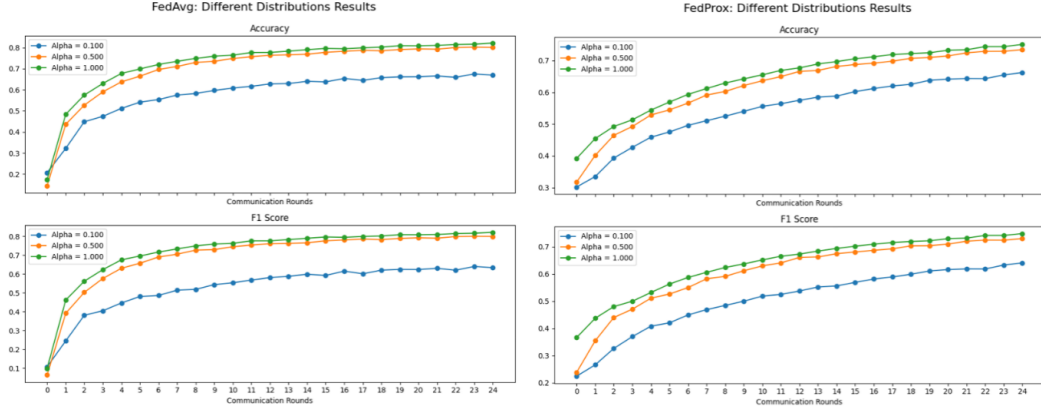


Figure 4: Accuracy and F1 score of FedAvg and FedProx on CIFAR-10 for Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$.

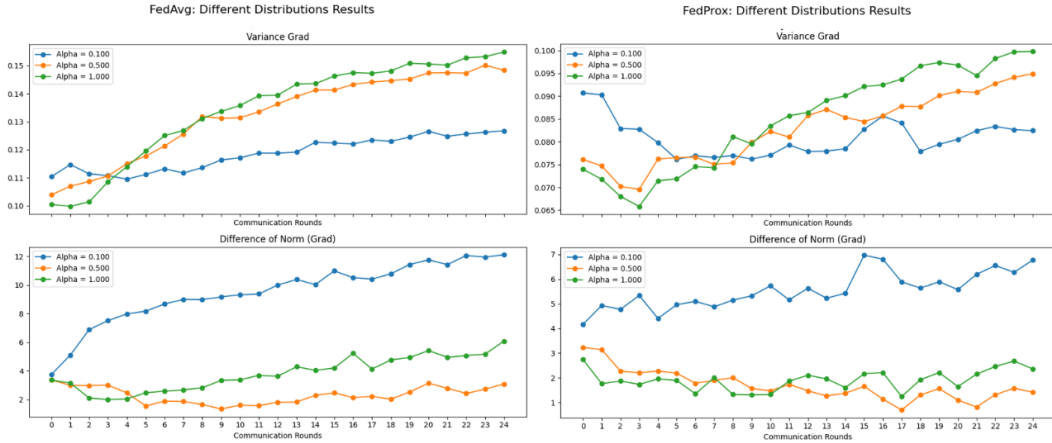


Figure 5: Variance of updates and difference of gradient norms for FedAvg and FedProx.

Toy problem: FEDSAM_HESSIAN on MNIST. With 5 clients, FEDSAM_HESSIAN replaces the first-order perturbation with the leading Hessian eigenvector, computed with PyHessian [Yao et al., 2020]. It outperforms FedSAM in test accuracy and converges faster across Dirichlet concentrations (Figure 6), but is approximately $10\times$ slower than FedSAM, even with power iteration limited to 10 iterations.

B Federated Setup Details

The federated scenario proceeds through communication rounds of (1) global model broadcast, (2) local training on a selected subset of clients, (3) communication of model updates and any auxiliary states required by the aggregation scheme, and (4) server aggregation (Figure 7a). We use

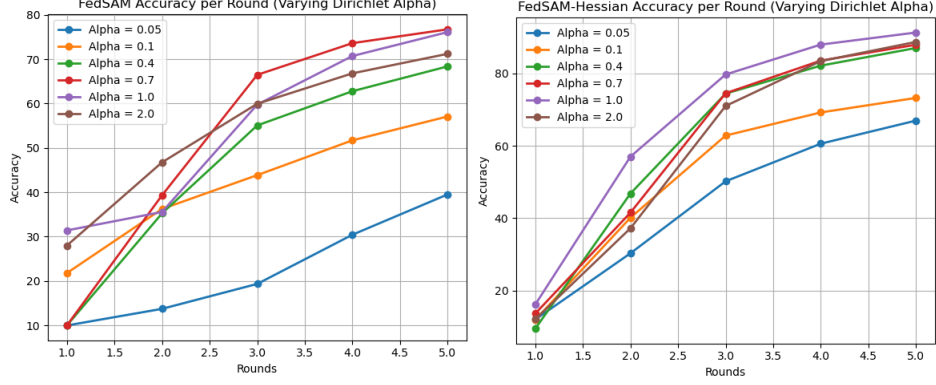


Figure 6: FedSAM vs. FEDSAM_HESSIAN on MNIST (5 clients) for varying Dirichlet α .

the centralized, server-based form of FL, as is common in practice. Figure 8 lists the states maintained by clients and server, Figure 7b illustrates client drift, and Figure 9 shows the model.

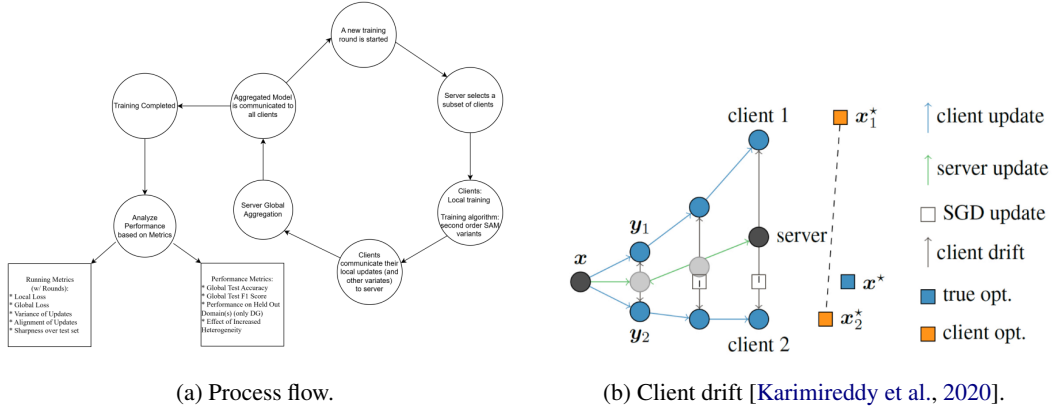


Figure 7: Federated training round and client drift under heterogeneity.

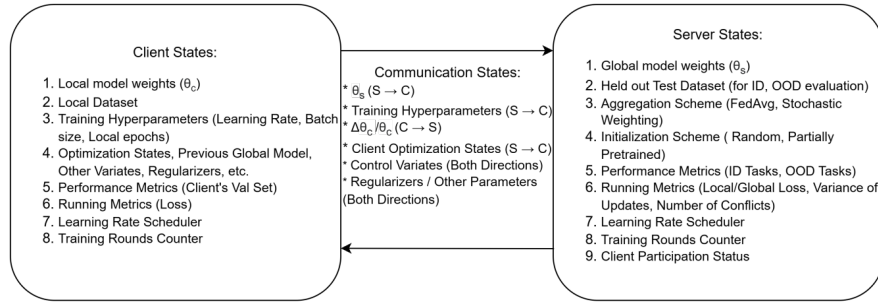


Figure 8: States maintained and communicated by clients and server.

C FedGloSS Integration

FedGloSS [Caldarola et al., 2024] optimizes the global sharpness-aware objective

$$F(w) = \frac{1}{|C|} \sum_{k \in C} F_k(w), \quad F_k(w) \triangleq \max_{\|\epsilon\| \leq \rho} f_k(w + \epsilon), \quad (8)$$

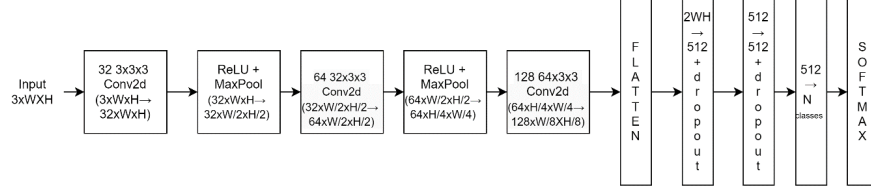


Figure 9: SimpleCNN architecture.

where C is the set of clients, F_k the sharpness-aware local loss, and f_k the local loss on client k . As the global gradient $\nabla F(w)$ is unavailable due to privacy constraints, the global perturbation is approximated from the previous round’s pseudo-gradient. Integrated with our client-side variants, one round proceeds as follows.

1. **Local optimization.** Each client applies a second-order FedSAM variant and sends only its updated parameters w_k .
2. **Server-side perturbation.** The server aggregates $w_t = \sum_{k \in C_t} \frac{N_k}{N} w_k$ and estimates the global perturbation and perturbed model

$$\tilde{\epsilon}^t(w) = \rho \frac{\Delta \tilde{w}^{t-1}}{\|\Delta \tilde{w}^{t-1}\|}, \quad \tilde{w}_t = w_t + \tilde{\epsilon}_t(w). \quad (9)$$

3. **Global update with ADMM.** Clients train from $\tilde{w}^t$; the resulting pseudo-gradient and global update are

$$\Delta \tilde{w}^t = \sum_{k \in C_t} \frac{N_k}{N} (\tilde{w}^t - \tilde{w}_k^t), \quad w^{t+1} = w^t - \eta_s \Delta \tilde{w}^t, \quad (10)$$

where η_s is the server learning rate. To keep the approximated perturbation consistent with the true one, FedGloSS applies ADMM to

$$\frac{1}{|C|} \sum_{k \in C} \left(f_k + \sigma_k^\top (w^t - w_k^t) + \frac{1}{2\beta} \|w^t - w_k^t\|_2^2 \right) \quad \text{s.t.} \quad w = w_k, \quad (11)$$

with Lagrange multipliers σ_k and hyperparameter β .

4. **Model distribution.** The updated global model w^{t+1} is broadcast for the next round.

D Codebase

The code is organized into `server/` and `client/` directories with a one-to-one correspondence between each server variant and its client counterpart (both inheriting from base `Server` and `Client` classes); `model/` (architectures); `utils/` (dataset loading and synthetic splitting, evaluation, and running metrics such as update variance and cosine similarity); and `results/`. All experiment state is encapsulated in a single `Args` object, so that an experiment reduces to:

```
args = Args(training_rounds=35, local_epochs=5, rule_arg=0.3, rho=0.01)
serve = FedSAM_Hessian(args)
serve.run_experiment() # results saved as JSON under results/
```